

KINGA Claims Manager Portal — Architecture Review v4.0

Document Classification: Internal Engineering Architecture Review

Portal Under Review: Claims Manager Portal (`/insurer/claims-manager`)

Review Version: 4.0 — Product Owner Response and Revised Target State

Prepared by: KINGA Platform Engineering

Date: June 2026

Status: Final

Purpose

This document is a structured response to the product owner architectural challenge raised against the v3.0 Alignment and Redesign Audit. It accepts the challenge where the reasoning is sound, pushes back where there is a legitimate counter-argument, and synthesises both perspectives into a revised target state architecture and implementation priority order.

The central challenge is this: **the v3.0 audit still designed the Claims Manager Portal as an analytics dashboard rather than an operational command centre**. That challenge is accepted. This document corrects that orientation.

Section 1 — Accepting the Core Architectural Challenge

The product owner's most important observation is that the v3.0 layout placed KPI cards in Row 1 and the intelligence bar in Row 2. That sequence reflects an analytics mindset: show the executive summary first, then the detail. A Claims Manager does not need an executive summary. A Claims Manager needs to know, within ten seconds of opening the portal, what requires immediate attention.

The correct mental model is not “dashboard” — it is “morning briefing.” When a Claims Manager opens the portal at the start of the day, the portal should answer six questions before they have to click anything:

1. What is stuck in my queues right now?
2. What is breaching SLA?
3. What decisions are waiting for me?
4. Which claims create risk if I ignore them today?
5. Who on my team is overloaded?
6. What is my backlog trajectory?

The v3.0 layout answered none of these questions above the fold. The revised layout must answer all six before the user reaches the tab structure.

This architectural reorientation is accepted in full.

Section 2 — Accepting the Specific Recommendations

2.1 Queue Health as Row 1

Accepted. The product owner is correct that Claims Managers live in queues. Queue health is the primary operational signal. Moving it to Row 1 is the correct architectural decision.

The v3.0 audit identified that `getDashboardStats.workflowStateCounts` is already fetched on every dashboard load and contains the queue count data. The revised Queue Health matrix adds Average Age and Oldest Claim (from `workflowAnalytics.getProcessingTimesByStage`) and SLA Breaches (from `workflowAnalytics.getBottlenecks`). All three data sources exist in the backend.

The revised Queue Health Row 1 specification is:

Queue Stage	Count	Average Age	Oldest Claim
Intake	<code>workflowStateCounts.intake_queue</code>	<code>getProcessingTimesByStage</code>	Derivable from <code>workflow_audit_trail</code>
Under Assessment	<code>workflowStateCounts.under_assessment</code>	<code>getProcessingTimesByStage</code>	Derivable
Internal Review	<code>workflowStateCounts.internal_review</code>	<code>getProcessingTimesByStage</code>	Derivable
Technical Approval	<code>workflowStateCounts.technical_approval</code>	<code>getProcessingTimesByStage</code>	Derivable
Financial Decision	<code>workflowStateCounts.financial_decision</code>	<code>getProcessingTimesByStage</code>	Derivable
Repair Assigned	<code>workflowStateCounts.repair_assigned</code>	<code>getProcessingTimesByStage</code>	Derivable

Engineering note: The “Oldest Claim” column requires a new query — a `MAX(TIMESTAMPDIFF(HOUR, entered_state_at, NOW()))` per stage. This is not currently returned by any existing procedure and requires a new `getQueueAgeDetails` procedure or an extension to `getProcessingTimesByStage`. This is a one-day engineering task.

2.2 Escalation Centre as Row 2

Accepted. The Escalation Centre should be the second thing a Claims Manager sees, not buried in an intelligence bar. The product owner’s expanded definition is adopted:

Escalation Category	Source	Backend Procedure
High Value Pending	Claims at <code>financial_decision</code> above threshold	<code>byStatus</code> + amount filter
High Fraud Risk	<code>fraudRiskLevel</code> high/critical/elevated	<code>getFraudAlerts</code>
Disputed	<code>workflowState = disputed</code>	<code>getEscalations</code>
Stuck Claims	Claims exceeding dwell threshold	<code>getBottlenecks</code>
SLA Breaches	Claims exceeding SLA per stage	<code>getSLACompliance</code>
Executive Overrides	<code>executiveOverride = 1</code> in <code>workflow_audit_trail</code>	New query required

The Escalation Centre must be actionable, not informational. Each category should be a clickable count that opens a filtered claim list. The Claims Manager must be able to act on any escalation category without navigating to a separate tab.

2.3 Approval Workbench as Row 3

Accepted. This is the most significant omission from v3.0. The most important responsibility of a Claims Manager is approvals, yet v3.0 buried approval visibility inside the Review Queue tab. A dedicated Approval Workbench panel above the tabs is the correct design.

The Approval Workbench specification:

Metric	Source	Action
Awaiting Technical Approval	<code>byStatus({ status: "technical_approval" }) count</code>	Click → Review Queue filtered to technical_approval
Awaiting Financial Decision	<code>byStatus({ status: "financial_decision" }) count</code>	Click → Review Queue filtered to financial_decision
High Value Pending	Claims above automation threshold at financial_decision	Click → filtered list
Oldest Approval	MAX age of claims at technical_approval or financial_decision	Click → sorted Review Queue
Average Approval Age	AVG age of claims at approval stages	Informational

Engineering note: The `byStatus` procedure currently fetches up to 200 claims per status. The Approval Workbench needs count and age aggregates, not full claim lists. A lightweight `getApprovalWorkbenchMetrics` procedure should be added to avoid over-fetching.

2.4 Capacity Forecasting

Accepted with scope clarification. The product owner's framing is precise: this is not AI forecasting, it is simple arithmetic. If intake rate exceeds completion rate, the backlog will grow. That calculation requires only two data points: 7-day intake trend and 7-day completion trend.

The `getManagerOverview` procedure already returns a `cycleTrend` array of daily claim counts. The completion trend requires a similar query on `closedAt` dates. Both are derivable from the existing `claims` table with no schema changes.

The Capacity Forecasting panel should show:

- Current Backlog (active claim count)
- 7-Day Intake (claims created in last 7 days)
- 7-Day Completions (claims closed in last 7 days)
- Backlog Trajectory (intake – completions, with directional indicator: growing / stable / shrinking)

This is a half-day engineering task. The directional indicator alone — a simple arrow showing whether the backlog is growing or shrinking — is a genuinely powerful operational signal that most claims systems do not surface.

2.5 Recovery Watchlist

Accepted. The product owner is correct that generic recovery counts (Open, Demand Sent, Settled) provide no actionable signal. A Claims Manager does not need to know how many recovery cases are open — they need to know which ones require attention today.

The Recovery Watchlist specification:

Category	Definition	Source
Recovery Eligible	Claims closed with third-party liability, no recovery case opened	Derivable from <code>claims</code> + <code>recovery_cases</code> join
Demand Outstanding	Recovery cases with <code>demandSent = true</code> and no response after threshold	<code>recovery_cases</code> table
Deadline Approaching	Recovery cases with <code>deadlineDate</code> within 7 days	<code>recovery_cases</code> table
High Value Recoveries	Recovery cases above value threshold	<code>recovery_cases.estimatedRecoveryAmount</code>

Engineering note: “Recovery Eligible” requires a join between `claims` (closed, with third-party involvement) and `recovery_cases` (no matching case). This is a new query. The other three categories are straightforward filters on the existing `recovery_cases` table.

2.6 Claims Requiring Attention Today — Management by Exception

Accepted as the highest-value new recommendation. The product owner's description of this widget is precise and operationally correct. A single "Attention Required" queue, built from exception rules, provides more operational value than most of the rest of the dashboard combined.

The exception rules proposed are:

Rule	Threshold	Source
SLA breach	Stage dwell exceeds threshold	<code>getBottlenecks</code>
Fraud score	<code>fraudRiskScore > 80</code>	<code>claims.fraudRiskScore</code>
High value pending	Claim above automation threshold at approval stage	<code>claims.totalClaimAmount + workflowState</code>
No update	<code>updatedAt > 7 days ago</code> and claim is active	<code>claims.updatedAt</code>
Multiple send-backs	More than 2 backward transitions in <code>workflow_audit_trail</code>	<code>workflow_audit_trail</code> count query
Executive override	<code>executiveOverride = 1</code> in recent <code>workflow_audit_trail</code>	<code>workflow_audit_trail</code>
Disputed	<code>workflowState = disputed</code>	<code>claims.workflowState</code>

The "Attention Required" widget should display a total count with a breakdown by rule category. Each category should be a clickable link that opens the filtered claim list. The widget should be positioned at the top of the Escalation Centre row, as it is the synthesis of all escalation signals into a single operational queue.

Engineering note: This requires a new `getAttentionRequired` procedure that runs all seven rule queries in parallel and returns a structured count object. The individual queries are all straightforward. The procedure should be designed to run efficiently — using `COUNT` aggregates rather than full row fetches — to avoid performance impact on dashboard load time.

Section 3 — Challenging One Recommendation

3.1 Fraud Funnel Removal

Partially challenged. The product owner argues that the Fraud Funnel (Flagged → Investigated → Confirmed) is useful for executives but not for operations. The counter-argument is that a Claims Manager does need to know whether fraud cases are progressing through investigation — not as a strategic metric, but as an operational queue health signal.

The product owner's alternative framing — "Fraud Cases Awaiting Action, Awaiting Risk Review, Delaying Approval, Older Than SLA" — is operationally superior to the funnel view for daily operations. This framing is adopted as the primary fraud visibility surface.

However, the Fraud Funnel is not removed entirely. It is relocated to the Reports Centre as a period-based report rather than a live dashboard widget. A Claims Manager reviewing monthly performance needs to understand whether the fraud investigation pipeline is healthy. That is a reporting function, not an operational monitoring function.

Resolution: Replace the Fraud Funnel dashboard widget with an **Operational Fraud Queue** showing the four categories the product owner specified. Retain the Fraud Funnel as a reportable metric in the Claims Operations Monthly Report.

The Operational Fraud Queue specification:

Category	Definition	Source
Fraud Cases Awaiting Action	High/critical fraud risk, no investigation comment in last 48 hours	<code>getFraudAlerts</code> + <code>workflow_audit_trail</code>
Fraud Cases Awaiting Risk Review	High fraud risk at <code>internal_review</code> or <code>technical_approval</code> stage	<code>getFraudAlerts</code> + <code>workflowState</code> filter
Fraud Cases Delaying Approval	High fraud risk at <code>financial_decision</code> stage	<code>getFraudAlerts</code> + <code>workflowState</code> filter
Fraud Cases Older Than SLA	High fraud risk claims exceeding stage dwell threshold	<code>getFraudAlerts</code> + <code>getBottlenecks</code>

Section 4 — Revised Target State Architecture

4.1 Design Philosophy

The revised architecture is built on a single principle: **the portal is a morning briefing, not a dashboard**. Every row answers a specific operational question. The Claims Manager should be able to answer all six core questions within ten seconds of opening the portal, before interacting with any tab or filter.

The KPI cards from v3.0 are not removed — they are relocated to a compact summary strip below the Workforce Intelligence row. They remain available for period-based context but are no longer the first thing the Claims Manager sees.

4.2 Revised Layout

Row 1 — Queue Health Matrix

The primary operational signal. Six queue stages displayed as a compact table with Count, Average Age, Oldest Claim, and SLA Breaches per stage. Each row is clickable and navigates to the Active Claims tab filtered by that workflow state.

Data sources: `getDashboardStats.workflowStateCounts`, `workflowAnalytics.getProcessingTimesByStage`, `workflowAnalytics.getBottlenecks`, new `getQueueAgeDetails` procedure.

Row 2 — Escalation Centre + Attention Required

Two panels side by side. The left panel is the “Attention Required” widget — a single count with breakdown by exception rule. The right panel is the Escalation Centre with six categories (High Value, High Fraud, Disputed, Stuck, SLA Breach, Executive Override). Both panels are fully actionable.

Data sources: New `getAttentionRequired` procedure, `getEscalations`, `getFraudAlerts`, `getBottlenecks`, `getSLACompliance`.

Row 3 — Approval Workbench + Capacity Forecasting

Two panels side by side. The left panel is the Approval Workbench (Awaiting Technical Approval, Awaiting Financial Decision, High Value Pending, Oldest Approval, Average Approval Age). The right panel is the Capacity Forecasting strip (Current Backlog, 7-Day Intake, 7-Day Completions, Backlog Trajectory indicator).

Data sources: New `getApprovalWorkbenchMetrics` procedure, `getManagerOverview.cycleTrend`, new completion trend query.

Row 4 — Workforce Intelligence

Three panels: Processor Performance, Assessor Performance, Workload Distribution. Collapsible to reduce visual weight when not in use.

Data sources: `analytics.getUserProductivity`, `analytics.getAssessorPerformance`.

Row 5 — Existing Tab Structure (preserved)

Intake Queue, Review Queue, Active Claims, Fraud Alerts (enhanced with Operational Fraud Queue), Processed Claims, Fleet Approvals. Report buttons added to Review Queue and Fraud Alerts tabs. No existing tabs removed.

Row 6 — Compact KPI Summary Strip

Total Claims, Completion Rate, Savings Identified, Average Cycle Days — displayed as a compact four-cell strip for period-based context. Not the primary operational surface.

Data sources: `getManagerOverview.kpis`.

4.3 Recovery Watchlist Placement

The Recovery Watchlist replaces the Recovery KPI row. It is positioned as a collapsible panel between Row 4 (Workforce Intelligence) and Row 5 (Tabs). It shows Recovery Eligible, Demand Outstanding, Deadline Approaching, and High Value Recoveries with counts and drill-down links to the Recovery Portal.

4.4 Architecture Comparison

Row	v3.0 Layout	v4.0 Layout	Change
1	KPI Cards	Queue Health Matrix	Replaced
2	Intelligence Bar (7 panels)	Escalation Centre + Attention Required	Replaced
3	Tabs	Approval Workbench + Capacity Forecasting	New
4	Workforce Intelligence (below tabs)	Workforce Intelligence	Moved above tabs
5	—	Tabs	Moved down
6	—	Compact KPI Strip	New (relocated)

Section 5 — Revised Implementation Priority Order

The product owner's revised priority order is adopted with minor additions for engineering dependencies.

Critical — Production Blockers (1.5 days)

Item	Rationale	Effort
Fix <code>closeForProcessing</code> procedure	Ambiguous audit trail; incorrect governance for closure decisions	0.5 days
Implement dedicated Escalation procedure	“Escalate” button currently misroutes claims via send-back	1 day

High — Operational Command Centre (5.5 days)

Item	Rationale	Backend Available	Effort
Queue Health Matrix (Row 1)	Primary operational signal; queue count data already fetched	Partial — needs <code>getQueueAgeDetails</code>	1.5 days
Escalation Centre + Attention Required (Row 2)	Morning briefing — exceptions surface immediately	Partial — needs <code>getAttentionRequired</code>	1.5 days
Approval Workbench (Row 3)	Most important responsibility; currently buried in tab	Partial — needs <code>getApprovalWorkbenchMetrics</code>	1 day
Claims Ageing panel	Answers “which stages are slowing down?”	<code>getProcessingTimesByStage</code> exists	0.5 days
Workflow Bottlenecks panel	Proactive bottleneck identification	<code>getBottlenecks</code> exists	0.5 days
Fleet Approvals sidebar navigation	Discoverability	Tab exists	0.25 days

Medium — Management Intelligence (5.5 days)

Item	Rationale	Backend Available	Effort
Workforce Intelligence (Row 4)	Claims Managers manage people, not claims	<code>getUserProductivity</code> , <code>getAssessorPerformance</code> exist	1.5 days
Send-back Analytics	Rework identification; training and quality signals	<code>workflow_audit_trail</code> data available	1 day
Recovery Watchlist	Actionable recovery oversight	Partial — needs eligibility query	1 day
Capacity Forecasting (Row 3)	Backlog trajectory — simple arithmetic	<code>getManagerOverview.cycleTrend</code> partial	0.5 days
Operational Fraud Queue (replace Fraud Funnel)	Operational fraud visibility vs strategic funnel	<code>getFraudAlerts</code> exists	0.5 days
Report Integration (Review Queue + Fraud Alerts)	11 authorised reports with no UI entry point	All report generators exist	1 day

Low — Refinements (2.25 days)

Item	Rationale	Effort
KPI Trend Analysis (compact strip)	Period-based context; not primary operational surface	0.5 days
Structured send-back reason capture	Enables rework analytics	0.5 days
Reopen capability for disputed claims	<code>closed</code> → <code>disputed</code> transition not exposed in UI	0.5 days
Audit metadata improvements	Record automation threshold at approval time	0.25 days
Merge Recently Closed card into Processed tab	Remove redundant UI section	0.25 days
Validate targetRole in send-back dialog	Clearer error messages	0.25 days

Revised Total Engineering Estimate

Priority	Items	Effort
Critical	2	1.5 days
High	6	5.5 days
Medium	6	5.5 days
Low	6	2.25 days
Total	20	14.75 days

The Critical + High items represent **7 days** of engineering work. At the end of those 7 days, the portal will function as a genuine Claims Operations Command Centre: queue health visible immediately, exceptions surfaced automatically, approvals in a dedicated workbench, and capacity trajectory visible at a glance.

Section 6 — New Procedures Required

The revised architecture requires five new backend procedures not identified in v3.0. All are derivable from existing data with no schema changes.

Procedure	Router	Query Description	Effort
<code>getQueueAgeDetails</code>	<code>workflowAnalytics</code>	MAX and AVG dwell time per workflow stage, plus oldest claim per stage	0.5 days
<code>getAttentionRequired</code>	<code>claims</code>	Parallel COUNT queries for all seven exception rules; returns structured count object	0.75 days
<code>getApprovalWorkbenchMetrics</code>	<code>claims</code>	COUNT and AVG age for claims at <code>technical_approval</code> and <code>financial_decision</code> ; oldest claim age	0.5 days
<code>getCapacityForecast</code>	<code>claims</code>	7-day intake count, 7-day completion count, current backlog, trajectory direction	0.5 days
<code>getRecoveryWatchlist</code>	<code>recovery</code>	Recovery Eligible (join query), Demand Outstanding, Deadline Approaching, High Value counts	0.75 days

Total new procedure engineering: **3 days** (included in the High and Medium estimates above).

Section 7 — Final Architectural Assessment

7.1 The Philosophical Shift

The product owner's challenge identifies a fundamental design error that persisted through v1.0, v2.0, and v3.0: the portal was designed by people thinking about what data to display rather than what decisions to support. An analytics dashboard asks “what data do we have?” An operational command centre asks “what decisions does this person make, and what do they need to make them well?”

A Claims Manager makes six categories of decisions every day: queue routing decisions, escalation decisions, approval decisions, workforce allocation decisions, exception management decisions, and capacity management decisions. Every row of the revised architecture maps to one of these decision categories.

7.2 What Makes This Architecture Genuinely Differentiated

Most claims administration systems provide claim lists with status filters. Some provide basic KPI dashboards. Very few provide a morning briefing that surfaces exceptions automatically, shows queue health at a glance, and gives the Claims Manager a single “Attention Required” queue that synthesises all operational risk signals.

The “Attention Required” widget in particular is the kind of feature that Claims Managers in insurers currently build manually — they maintain spreadsheets of claims that need attention, updated each morning. Automating that process and surfacing it as a live widget is a meaningful product differentiator.

7.3 Revised Production Readiness Verdict

Score: 5.8 / 10 — Revised to 4.5 / 10 against the v4.0 standard

The score is revised downward because the v4.0 standard is more demanding than the v3.0 standard. The portal is being assessed not against “does it have analytics capability” but against “does it function as a Claims Operations Command Centre.” Against that standard, the current portal — which has no Queue Health matrix, no Attention Required queue, no Approval Workbench, and no Capacity Forecasting — scores lower.

The path to a command centre is now precisely defined. The Critical + High items (7 days) deliver the core command centre capability. The Medium items (5.5 days) add management intelligence. The Low items (2.25 days) refine the experience.

Recommended Go-Live Condition (revised): Complete all Critical and High items before production deployment. The portal must surface Queue Health, Escalation Centre, Approval Workbench, and Attention Required before it is presented to Claims Managers as their primary operational tool. These are not enhancements — they are the minimum viable command centre.

Appendix — v3.0 to v4.0 Recommendation Disposition

v3.0 Recommendation	v4.0 Disposition	Change
Fix <code>closeForProcessing</code>	Retained — Critical	No change
Dedicated Escalation procedure	Retained — Critical	No change
Queue Health panel (Row 2)	Promoted to Row 1 with expanded spec	Promoted
Escalation Centre panel (Row 2)	Promoted to Row 2 with expanded categories	Promoted + expanded
Approval Workbench	New — added to Row 3	New
Claims Ageing panel	Retained — High	No change
Workflow Bottlenecks panel	Retained — High	No change
Fleet Approvals navigation	Retained — High	No change
Workforce Intelligence (Row 4)	Moved above tabs — Row 4	Moved
Send-back Analytics	Retained — Medium	No change
Recovery Watchlist	Replaces Recovery KPI row	Upgraded
Capacity Forecasting	New — added to Row 3	New
Operational Fraud Queue	Replaces Fraud Funnel widget	Replaced
Report Integration	Retained — Medium	No change
KPI Cards (Row 1)	Demoted to compact strip (Row 6)	Demoted
Fraud Funnel widget	Moved to Reports Centre	Relocated
Attention Required queue	New — highest-value new feature	New